ANN8: Hopfield Network (4 Vectors)

Cell 1: Define 4 Patterns

Code:

```
X = np.array([...4 patterns...])
```

Explanation (what to say in viva):
Four patterns are stored.

Increasing patterns increases complexity.

May cause interference.

Cell 2: Weight Matrix

Code:

```
W = np.zeros((n,n))
for p in X:
    W += np.outer(p,p)
```

Explanation (what to say in viva):
Stores all patterns.

More patterns → more interference.

Cell 3: Diagonal Zero

Code:

```
np.fill_diagonal(W,0)
```

Explanation (what to say in viva):
Removes self-connection.

Important for stability.

Cell 4: Recall

Code:

```
x = sign(np.dot(W,x))
```

Explanation (what to say in viva):
Network updates state.

Tries to converge to nearest stored pattern.

Cell 5: Noisy Input

Code:

```
noisy = [...]
```

Explanation (what to say in viva):
Testing network with noisy input.

Checks memory recall ability.

Cell 6: Output

Code:

```
print(x)
```

Explanation (what to say in viva):

Shows recovered pattern.

May not always be correct due to capacity limits.